

Treatment of Unwanted Hair Growth

The use of anti-androgens may help reduce hair growth, but monotherapy is unlikely to eliminate it completely (Box 86-3). Temporary hair removal methods such as shaving, plucking, and depilatory creams can be irritating and painful. Topical eflornithine 13.9%—an inhibitor of ornithine decarboxylase—is approved for reducing unwanted facial hair in women when applied twice daily.

Laser epilation is less permanent than commonly believed and works best with repeated treatments. Patients should be counseled against unrealistic expectations. The mechanism involves selective photothermolysis, where melanin in the hair shaft absorbs light energy, leading to targeted destruction of hair follicles. Combining topical eflornithine with laser therapy may enhance outcomes by slowing regrowth after hair removal.

Weight loss can be beneficial in obese individuals, as obesity lowers sex hormone-binding globulin (SHBG) levels, increasing free testosterone, and contributes to insulin resistance, both of which can exacerbate hirsutism.

Systemic treatments show marginal differences in efficacy, but all typically require 6 to 12 months to achieve clinically significant miniaturization of terminal hairs. Options include: - **Oral anti-androgens** such as spironolactone (100–200 mg daily) or cyproterone acetate - **Oral contraceptives**, which suppress androgen synthesis by reducing gonadotropin secretion; they are most effective when combined with anti-androgens - **Finasteride**, a 5 α -reductase inhibitor

In cases of congenital adrenal hyperplasia, low-dose dexamethasone can be used to suppress adrenal androgen overproduction. Hyperprolactinemia may be managed medically (e.g., bromocriptine) or surgically, depending on the cause, and may also require adjunctive anti-androgen therapy.

Hypertrichosis

Congenital Hypertrichosis Lanuginosa

Congenital hypertrichosis lanuginosa is characterized by a generalized, confluent overgrowth of fine, silvery blonde to gray lanugo hair present at birth or in early infancy. It is extremely rare (approximately 1 in 1 billion) and typically follows an autosomal dominant inheritance pattern with variable expressivity. Most affected children are otherwise healthy, although dental anomalies may occur. The hair distribution may persist, increase, or regress with age.

Other Congenital Disorders with Hypertrichosis

Other congenital conditions associated with generalized hypertrichosis exist, but none exhibit the uniform distribution seen in congenital hypertrichosis lanuginosa (Table 86-10). These include:

- **Ambras syndrome (hypertrichosis universalis congenita):** Autosomal dominant condition featuring long, thick hair, especially on the face, ears, and shoulders. It is often accompanied by facial dysmorphism and dental abnormalities.
- **X-linked dominant congenital generalized hypertrichosis:** Seen in a five-generation family, with more pronounced terminal hair on the face and upper body, particularly in males.
- **Gingival fibromatosis with hypertrichosis:** Autosomal dominant (rarely recessive) disorder associated with hypertrichosis of the face, eyebrows, limbs, and upper back, along with seizures and intellectual disability (oligophrenia). Hypertrichosis may appear at puberty, while gingival overgrowth typically emerges with primary tooth eruption.

Acquired Generalized Hypertrichosis

The sudden onset of generalized hypertrichosis without an identifiable drug trigger is a concerning clinical sign. Acquired hypertrichosis lanuginosa is strongly associated with underlying malignancy and may precede the diagnosis

of cancer. It should prompt a thorough systemic evaluation for occult neoplasm (Fig. 86-29; see also eFig. 86-29.1 in online edition; Table 86-10; Chap. [reference incomplete]).